

HIF1A contributes to the survival of aneuploid and mosaic pre-implantation embryos

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eLife Assessment

Sanchez-Vasquez et al establish an innovative approach to induce aneuploidy in preimplantation embryos. This **important** study extends the author's previous publications evaluating the consequences of aneuploidy in the mammalian embryo. In this work, the authors investigate the developmental potential of aneuploid embryos and characterize changes in gene expression profiles under normoxic and hypoxic culture conditions. Using a **solid** methodology they identify sensitivity to Hif1alpha loss in aneuploid embryos, and in further **convincing** experiments they assess how levels of DNA damage and DNA repair are altered under hypoxic and normoxic conditions.

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Abstract

Human fertility is suboptimal, partly due to error-prone divisions in early cleavage-stages that result in aneuploidy. Most human pre-implantation are mosaics of euploid and aneuploid cells, however, mosaic embryos with a low proportion of aneuploid cells have a similar likelihood of developing to term as fully euploid embryos. How embryos manage aneuploidy during development is poorly understood. This knowledge is crucial for improving fertility treatments and reducing developmental defects. To explore these mechanisms, we established a new mouse model of chromosome mosaicism to study the fate of aneuploid cells during pre-implantation development. We previously used the Mps1 inhibitor reversine to generate aneuploidy in embryos. Here, we found that treatment with the more specific Mps1 inhibitor AZ3146 induced chromosome segregation defects in pre-implantation embryos, similar to reversine. However, AZ3146-treated embryos showed a higher developmental potential than reversine-treated embryos. Unlike reversine-treated embryos, AZ3146-treated embryos exhibited transient upregulation of Hypoxia Inducible-Factor-1A (HIF1A) and lacked p53 upregulation. Pre-implantation embryos develop in a hypoxic environment *in vivo*, and hypoxia exposure *in vitro* reduced DNA damage in response to Mps1 inhibition and increased the proportion of euploid cells in the mosaic epiblast. Inhibiting HIF1A in mosaic embryos also decreased the proportion of aneuploid

cells in mosaic embryos. Our work illuminates potential strategies to improve the developmental potential of mosaic embryos.

Introduction

Humans exhibit suboptimal fertility relative to other mammals (Palmerola et al., 2022a). Only ~30% of human conceptions progress to live birth (Capalbo et al., 2021). Intriguingly, the first zygotic cleavage-stage divisions in human embryos are error prone, generally resulting in aneuploidy, in which cells gain or lose chromosomes (Allais & FitzHarris, 2022; Currie et al., 2022; Palmerola et al., 2022b; Pauerova et al., 2020; Vanneste et al., 2009). A high incidence of aneuploidy in the early cleavage divisions is thought to underlie low human fecundity and most developmental defects (Palmerola et al., 2022a). Both *in vivo* and *in vitro* fertilization (IVF) often give rise to mosaic embryos of diploid and aneuploid cells. Although it is predicted that ~60% of pre-implantation IVF embryos exhibit diploid-aneuploid mosaicism (Capalbo et al., 2021), our understanding of the embryo's ability to cope with such abnormalities is limited.

The incidence of aneuploidy declines at later stages of development (Shahbazi et al., 2020), but the mechanism underlying this decline is poorly understood. Human mosaic embryos can develop to term (Capalbo et al., 2021; Greco et al., 2015; Starostik et al., 2020). Specifically, implantation of low- and medium-grade mosaic embryos, defined as having 20-30% and 30-50% aneuploid extra-embryonic trophoblast cells in a biopsy, has been reported to be as likely to result in the birth of healthy babies as implantation of fully euploid embryos (Capalbo et al., 2021). Given that some mosaic embryos are able to develop to term, it is important to uncover the mechanisms that confer their viability.

Mouse models of chromosome mosaicism enable analyses that are not allowed with human embryos for ethical reasons. Mouse and human pre-implantation development are very similar to each other: both undertake cleavage divisions, compaction, blastocyst cavity formation and hatching, albeit with slightly different timings in the two species (Mole et al., 2020; Zhu et al., 2021). During pre-implantation development, cells on the outside of the embryo form the extra-embryonic trophoblast (TE) whereas cells on the inside form the pluripotent inner cell mass (ICM). Subsequently, cells of the ICM are segregated into the embryonic epiblast (EPI) and extra-embryonic primitive endoderm (PE). The TE will form the placenta, the PE will form the yolk sac and the EPI will form the fetus (Zhu & Zernicka-Goetz, 2020).

Recently, our group developed the first mouse model of chromosome mosaicism, based on the generation of aneuploidies induced by a drug reversine (Bolton et al., 2016; Singla et al., 2020). Reversine is a pan-Aurora kinase inhibitor that also antagonizes the A3 adenosine receptor and inhibits mitotic kinase monopolar spindle 1 (MPS1) (D'Alise et al., 2008; Santaguida et al., 2010). Our analysis revealed that reversine-treated mosaic embryos are aneuploid and overexpress p53 (Bolton et al., 2016). Importantly, reversine-treated cells in mosaic embryos are gradually eliminated from the EPI, particularly around the time of blastocyst implantation. As with human mosaic embryos (Capalbo et al., 2021), we found that murine mosaic embryos with at least 50% of euploid cells had a similar developmental potential as fully euploid embryos (Bolton et al., 2016; Singla et al., 2020).

As reversine affects p53 and may compromise cellular fitness (D'Alise et al., 2008; Santaguida et al., 2010), we sought to generate a complementary aneuploid model. To this end, we used the more specific Mps1 inhibitor, AZ3146 (Hewitt et al., 2010). Although AZ3146 and reversine both interfere with the spindle assembly checkpoint (Hewitt et al., 2010), they bind to Mps1 in a different manner (Lan & Cleveland, 2010). In mouse embryos, AZ3146 treatment was shown to double the occurrence of micronuclei, a marker of chromosome segregation defects, but without

negatively affecting cellular fitness (Vazquez-Diez et al., 2019). In the present study, we used AZ3146- and reversine-treated pre-implantation mouse embryos to dissect the mechanisms underlying aneuploid cell elimination and survival.

Results

AZ3146 treatment induces chromosome segregation defects in pre-implantation mouse embryos

To generate distinct models of aneuploidy, we treated 4- to 8-cell stage mouse embryos with AZ3146 (20 μ M) (Vazquez-Diez et al., 2019), as well as with reversine (0.5 μ M) (Bolton et al., 2016) as a positive control, or DMSO (vehicle) as a negative control (Fig. 1A). We evaluated how the different Mps1 inhibitors affect chromosome segregation by detecting nuclei and kinetochores in 8-cell embryos and counted chromosomes *in situ* (Pauerova et al., 2020). Micronuclei were identified as small DAPI-stained chromosomes that were clearly distinct from the nuclei (Fig. S1A). We examined 32 DMSO-treated control embryos and observed only 22 cells with micronuclei out of a total of 256 cells (Fig. S1B). In contrast, we observed 82 cells with micronuclei in a total of 144 individual cells from 18 reversine-treated embryos, and 182 cells in a total of 304 individual cells from 38 AZ3146-treated embryos (Fig. S1A). After normalization, the average of aneuploidy in DMSO-treated 8-cell embryos was 12.5%, compared to 75% in the reversine-treated embryos and 62.5% in the AZ3146-treated embryos (Fig. S1B). We also detected non-dividing nuclei as rounded circles with distinct DAPI intensities in embryos treated with reversine and AZ3146. Specifically, we observed 28 non-dividing cells (19%) in reversine-treated embryos and 24 non-dividing cells (7.9%) in AZ3146-treated blastomeres, but none in DMSO-treated blastomeres (Fig. S1B). The increased frequencies of micronuclei and non-dividing cells in response to Mps1 inhibition are consistent with elevated aneuploidy and chromosomal instability (Daughtry and Chavez, 2016; Vazquez-Diez et al., 2016), as we showed previously for reversine-treated embryos (Bolton et al., 2016).

Given these distinct responses to reversine and AZ3146, we examined how these treatments impact the lineages in the blastocyst. To this end, we treated embryos with DMSO, reversine or AZ3146 from the 4- to 8-cell stage, and then washed and cultured them until the late blastocyst stage (E4.5). We assessed lineages by immunofluorescence (IF) to detect the TE marker CDX2, the EPI marker NANOG, and the PE marker SOX17. We found that all lineages segregated normally, and that blastocyst morphology was similar in all three conditions (Fig. 1B). We quantified the number of cells in the blastocysts and found that DMSO-treated controls had a median of 93 cells in total, whereas reversine- and AZ3146-treated embryos had a median of only 79 and 82 cells, respectively (** P <0.0004, * P <0.05, Mann–Whitney U-test, n =28 embryos per treatment collected from three independent experiments) (Fig. 1C). DMSO-treated blastocysts contained a median of 15 EPI, 11 PE, and 67 TE cells, whereas reversine-treated blastocysts contained 12 EPI, 8 PE and 59 TE cells, and AZ3146-treated blastocysts had 11 EPI, 7 PE and 64 TE cells (Fig. 1C). These data suggest that reversine treatment compromises the development of all three lineages, as observed previously (Bolton et al., 2016), whereas AZ3146 mainly compromises ICM (EPI and PE) development.

To compare the developmental potential of blastocysts that had been treated with AZ3146 or reversine, we transferred them into opposite uterine horns of the same mouse and counted decidia, which reflect successful implantation, as well as viable embryos at E9.5. Although two (8%) reversine-treated blastocysts developed decidia (Fig. S1C), none gave rise to viable E9.5 embryos. In contrast, seven (30%) AZ3146-treated blastocysts developed decidia and five (21%) generated viable E9.5 embryos (Fig. S1D). Thus, AZ3146-treated embryos appear to have a higher developmental potential than reversine-treated embryos.

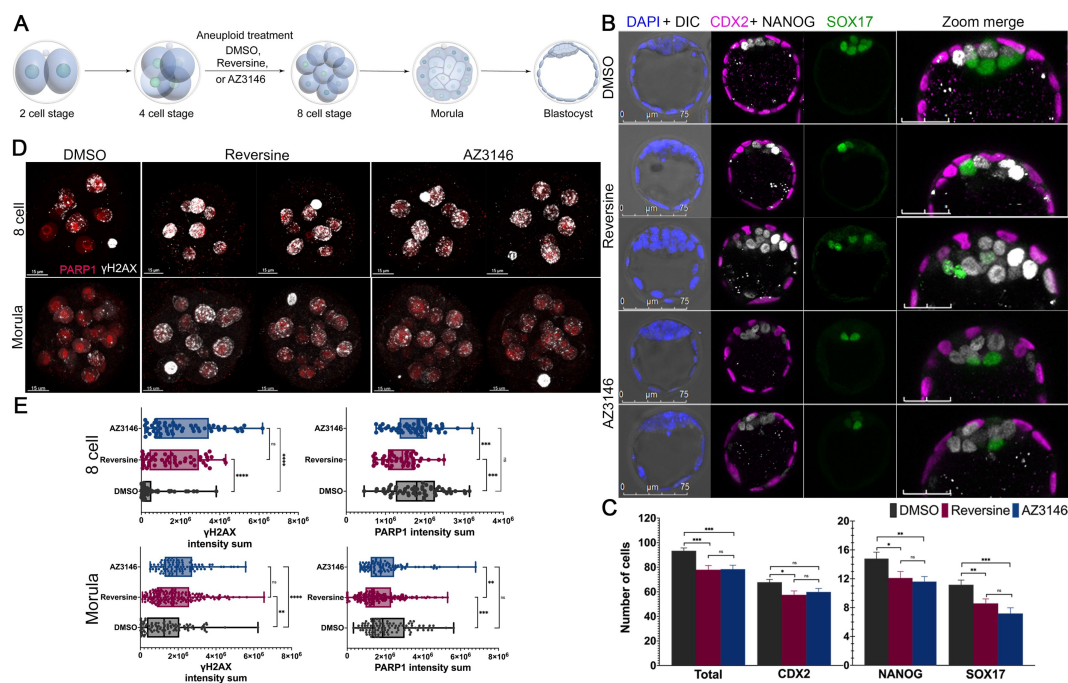


Figure 1

Lineage analysis of aneuploid embryos generated by selective Mps1 inhibitors AZ3146 and reversine.

(A) Graphic representation of 4-cell embryos treated with DMSO (control) or Mps1 inhibitors reversine (0.5 μ M) and AZ3146 (20 μ M) to inactivate the spindle assembly checkpoint (SAC) and induce chromosome segregation errors. After washing, embryos were cultured to the mature blastocyst stage (E4.5) and analyzed for lineage specification. (B) Immunofluorescence imaging of well-known lineage markers CDX2 (TE), NANOG (EPI) and SOX17 (PE) reveals that overall embryonic morphology and cavitation is not affected by Mps1 inhibition. (C) Number of cells in each lineage was quantified to evaluate the effect of drug treatment on blastocyst development. Importantly, both reversine and AZ3146-treatments reduce the number of cells in the ICM, marked by NANOG (EPI) and SOX17 (PE). Whereas the TE, marked by CDX2, is reduced only in the reversine-treated embryos. (n=28 embryos per treatment, cumulative of three independent experiments) (***P<0.0002, **P<0.01, *P<0.05, Mann-Whitney U-test, error bars represent s.e.m). (D) Analysis of DNA damage and DNA repair based on immunofluorescence against γ H2A.X (phosphorylated form of H2A.X, white) and PARP1 (red), respectively. (E) Intensity analysis shows that reversine and AZ3146 increase DNA damage at the 8-cell stage through morula stage compared with DMSO-treated embryos. Importantly, reversine appears to downregulate PARP1 expression at the 8-cell stage, which extends to morula stage embryos (***P<0.0004, **P<0.01, *P<0.05, t-test, error bars represent s.e.m).

Considering that cellular fitness and aneuploid stress are related to DNA damage and DNA repair, we first sought to understand if these parameters were affected in our drug treatments. We used IF to quantify the levels of the DNA repair marker PARP1 and of the DNA damage marker, phosphorylated H2A.X (γ H2A.X) at the 8-cell and morula stages (**Fig. 1D**). We observed elevated γ H2A.X levels in reversine- and AZ3146- treated 8-cell embryos compared to controls (**Fig. 1E**), which eventually returned to normal in AZ3146- treated blastocysts, but not in reversine-treated blastocysts (**Fig. S2A**).

PARP1 was specifically enriched in the EPI lineage in normal blastocysts, as assessed by IF and re-analysis of published scRNA-seq data (Deng et al., 2014) (**Fig. S2B**). Intriguingly, PARP1 levels were reduced at the 8-cell and morula stages in reversine-treated embryos compared to controls and to AZ3146-treated embryos (**Fig. 1D-E**). Moreover, late morula stage embryos treated with the PARP inhibitor Olaparib (10 μ M, Hou et al., 2022; Prasad et al., 2017) developed into smaller blastocysts with only 78 total cells, reflecting a relatively low number of ICM cells, only 8 EPI and 4 PE cells in DMSO-treated embryos (**Fig. S2C-E**). Treatment with reversine, but not AZ3146, further reduced the number of cells in the EPI lineage of Olaparib-treated embryos (**Fig. S2E**). Overall, these data suggest that PARP1 is required for proper development of the ICM, and that its reduced levels after reversine treatment may negatively impact development.

Reversine and AZ3146 activate distinct stress response pathways in pre-implantation embryos

Chromosome mis-segregation and aneuploidy are associated with different cellular stress pathways (Zhu et al., 2018). For instance, p53 is frequently activated following DNA damage, which can limit proliferation and trigger apoptosis (Abuetab et al., 2022; Soto et al., 2017; Thompson & Compton, 2010). The p38 mitogen-activated protein kinase (MAPK) can also be activated following DNA damage (Thompson and Compton, 2010). In aneuploid cells, p38 promotes apoptosis by inhibiting the transcription factor Hif-1 α (Simoes-Sousa et al., 2018), which otherwise promotes cell survival, proliferation, and metabolic changes in aneuploid cells and in response to hypoxia in different contexts (Hu et al., 2003; Simoes-Sousa et al., 2018).

To investigate how treatment with reversine or AZ3146 affect p53 and Hif1 α expression in mouse embryos, we performed RT-qPCR at the morula and blastocyst stages. We normalized to *Ppia* (Peptidylprolyl Isomerase A) mRNA, which is a stable reference gene in diploid and polyploid embryos (Gu et al., 2014). Reversine-treated embryos displayed a significant increase in p53 transcript levels at the morula (7-fold) and blastocyst (1.3-fold) stages compared with DMSO-treated embryos (** $P < 0.01$, * $P < 0.05$, Welch's t-test) (**Fig. 2A**), as we showed previously (Singla et al., 2020). In addition, reversine-treated embryos showed reduced expression of Hif1 α at the blastocyst stage, which would be consistent with p38 activation. In contrast, AZ3146-treated embryos did not show upregulation of p53 at either the morula or blastocyst stage (**Fig. 2A**). Moreover, AZ3146-treated embryos showed a transient increase in Hif1 α mRNA levels (3-fold) at the morula stage compared to DMSO- and reversine-treated embryos (**Fig. 2A**), which returned to normal levels at the blastocyst stage. HIF1A protein was present in DMSO- and reversine-treated embryos and elevated at the morula and blastocyst stages in AZ3146-treated embryos (**** $P < 0.0001$, ** $P < 0.01$, * $P < 0.05$, Mann-Whitney U-test) (**Fig. 2B**). HIF1A appeared to be mostly nuclear in morula, but mostly cytoplasmic in blastocysts, under all three conditions (**Fig. 2B**). Overall, these data suggest that treatment with reversine, but not AZ3146, induces multiple stress pathways in pre-implantation embryos. These differences may contribute to the increased developmental potential of embryos treated with AZ3146 versus reversine.

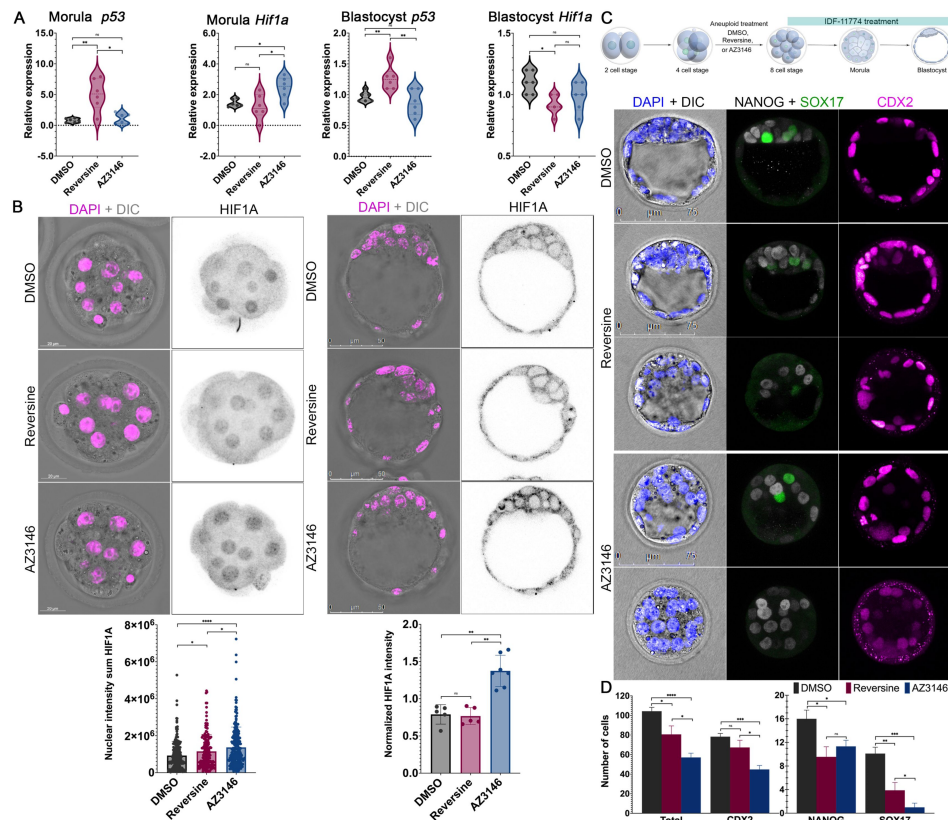


Figure 2

AZ3146-treated embryos elevate HIF1A activity to support formation of the TE and PE.

A) qPCR analysis of *p53* and *Hif1a* mRNA expression at morula and blastocyst stages reveals that *p53* is upregulated in reversine-treated embryos and that *Hif1a* is upregulated in AZ3146-treated embryos at morula stages; 3 biological replicates and 2 technical replicates per experiment with each replicate having a minimum of 16 embryos (** $P < 0.01$, * $P < 0.05$, Welch's t-test). B) Immunofluorescence against HIF1A (black) shows an increase in nuclear intensity in AZ3146-treated embryos at morula stages. At blastocyst stage, nuclear and cytoplasmic HIF1A are increased in AZ3146-treated embryos; normalization was based on DAPI staining (magenta). (**** $P < 0.0001$, ** $P < 0.01$, * $P < 0.05$, Mann-Whitney U-test, error bars represent s.e.m) (C) Graphic representation of 4-cell embryos treated with DMSO and aneuploid drugs. Downregulation of HIF1A was achieved by treatment with IDF-11774 immediately after wash of the aneuploid drugs. Immunofluorescence analysis of lineage specification in blastocyst cultured with the HIF1A inhibitor IDF-11774 using antibodies against CDX2 (TE), NANOG (EPI) and SOX17 (PE). Importantly, IDF-11774 appears to affect cavitation of some AZ3146-treated embryos. (D) Lineage analysis at the blastocyst stage shows that TE and PE specification are affected by IDF-11774 treatment. Number of cells in each lineage was quantified to evaluate the effect on blastocyst development. (**** $P < 0.0001$ ** $P < 0.01$, * $P < 0.05$, Mann-Whitney U-test, error bars represent s.e.m. $n = 20$ embryos per treatment collected from 3 independent experiments).

HIF1A activity is required for proper blastocyst formation after Mps1 inhibition

It was previously shown that *Hif1a*^{-/-} embryos undergo developmental arrest and lethality by E11 (Iyer et al., 1998 [\[1\]](#)). To assess the role of HIF1A in the embryo's response to reversine and AZ3146, we used a pharmacological approach to inhibit its function. Briefly, we tested two different small molecules that have been shown to inhibit HIF1A activity, PX-478 (Lee & Kim, 2011 [\[2\]](#); Zhao et al., 2015 [\[3\]](#)) and IDF-11774 (Ban et al., 2017 [\[4\]](#)). We treated control zygotes with DMSO, PX-478 (2 μ M), or IDF-11774 (20 μ M) until the blastocyst stage (**Fig. S3A** [\[5\]](#)). Unlike PX-478, IDF-11774 treatment did not significantly affect the total number of cells in the TE or the whole blastocyst compared to the control (**** $P < 0.0001$, Mann–Whitney U-test) (**Fig. S3B** [\[6\]](#)). Therefore, we used IDF-11774 to inhibit HIF1A in subsequent experiments.

Next, we treated embryos with DMSO, reversine, or AZ3146 from the 4-to 8-cell stage, washed them, and then inhibited HIF1A with IDF-11774 from the 8-cell to blastocyst stage (**Fig. 2C** [\[7\]](#)). Inhibition of HIF1A did not abolish cavitation but dramatically lowered the number of TE and especially PE cells in AZ3146-treated embryos compared to DMSO-treated controls (**Fig. 2D** [\[8\]](#)) and compared to AZ3146-treated embryos not exposed to IDF-11884 (**Fig. 1C** [\[9\]](#)). Inhibition of HIF1A also lowered the number of cells in reversine-treated embryos compared to controls, but the effect was smaller (**Fig. 2D** [\[10\]](#)). Overall, our data suggest that elevated HIF1A activity from the 8-cell stage onwards is particularly important to promote the survival of aneuploid TE and PE cells after AZ3146 treatment.

Taken together, our data suggest that AZ3146 and reversine have distinct effects on the pre-implantation embryo. Compared to reversine-treated embryos, AZ3146-treated embryos appear to have increased developmental potential, lack upregulation of p53, and show transient upregulation of *Hif1a*. Moreover, AZ3146-treated embryos have a greater dependence on HIF1A activity to form the TE and PE.

Hypoxia exposure attenuates DNA damage and blastomere defects in response to Mps1 inhibition

Pre-implantation development occurs in a hypoxic environment, and many clinics culture human embryos under hypoxic conditions (5%) (Houghton, 2021 [\[11\]](#)). It was reported that physiologic oxygen concentrations can improve the yield and quality of mammalian blastocysts (Ciray et al., 2009 [\[12\]](#); Nguyen et al., 2020 [\[13\]](#)) and increase the nuclear translocation of HIF1A in mouse blastocysts (Choi et al., 2021 [\[14\]](#)). To investigate how hypoxia affects the development of euploid and aneuploid mouse embryos, we repeated our experiments performed under normoxia (air, 20% O₂/5% CO₂; control) (**Fig. 1A** [\[15\]](#)) in hypoxic (5% O₂/5% CO₂) conditions (**Fig. 3A** [\[16\]](#)). We cultured embryos from the 2-cell stage until the blastocyst stage, treating them with DMSO, reversine, or AZ3146 from the 4-8 cell stage.

First, we examined lineage specification by quantifying the cell numbers in all three lineages at the blastocyst stage. We found that DMSO-treated blastocysts cultured under hypoxia had a median of 74 cells, representing 14 EPI cells, 6 PE cells and 54 TE cells (**Fig. 3B-C** [\[17\]](#)). These data suggest that, under our conditions, the cell number in the pre-implantation mouse blastocyst is reduced when cultured in hypoxia compared to normoxia (**Fig. 3C** [\[18\]](#), **1C** [\[19\]](#)), particularly in the TE and PE. Notably, reversine and AZ3146 treatment did not further reduce cell numbers for embryos cultured under hypoxia, except for in the PE (**Fig. 3C** [\[20\]](#)).

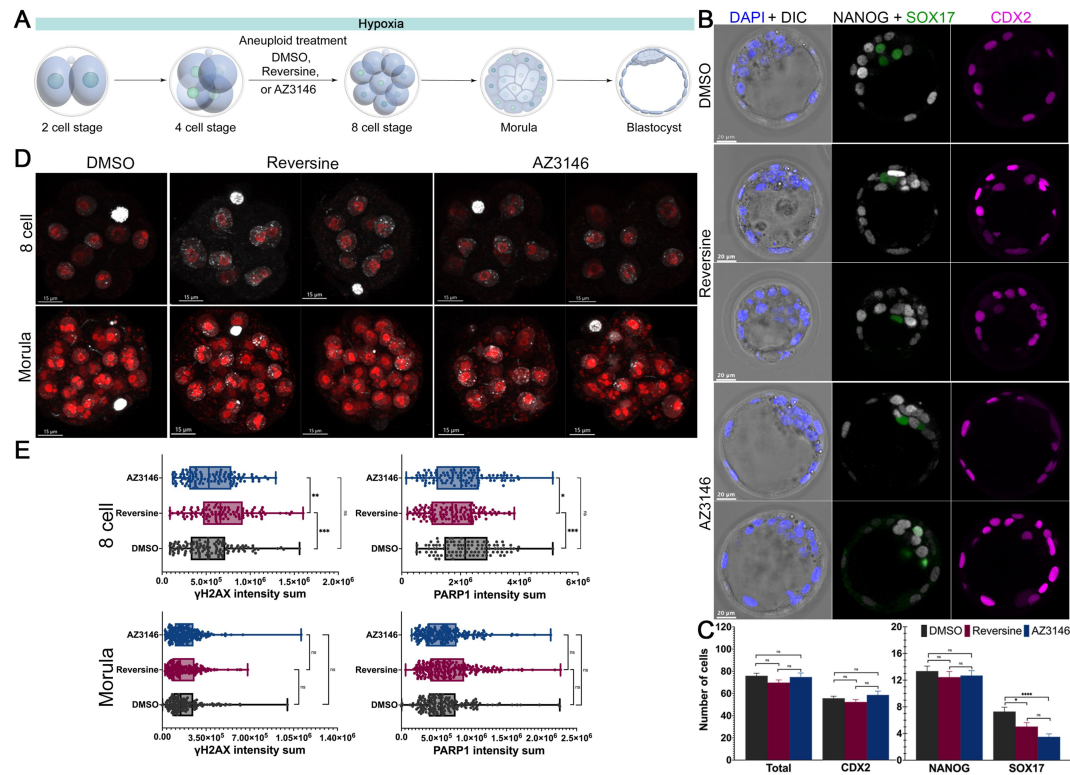


Figure 3

Hypoxia exposure reduces DNA damage and affects lineage proportions in the aneuploid blastocyst.

(A) Graphic representation of the hypoxia experiments. 2-cell embryos were cultured until the blastocyst stage in hypoxia conditions (5% oxygen). As before, 4-cell stage embryos were treated with DMSO or Mps1 inhibitors reversine and AZ3146 until the 8-cell stage. After washing, embryos were cultured to the mature blastocyst stage (E4.5) and analyzed for lineage specification. (B) Immunofluorescence imaging of well-known lineage markers CDX2 (TE), NANOG (EPI) and SOX17 (PE) reveals that overall embryonic morphology and cavitation is not affected by Mps1 inhibition or hypoxia. (C) Lineage analysis at blastocyst stage. Number of cells in each lineage was used to evaluate the effects in blastocyst development. (**** $P < 0.0001$, * $P < 0.05$, Mann-Whitney U-test, error bars represent s.e.m, $n = 18$ embryos per treatment collected from 3 independent experiments). (D) Immunofluorescence against PARP1 (red) and yH2A.X (white) in blastocyst after drug treatments. (E) Intensity analysis shows that, under hypoxia, at 8-cell stage, DNA damage is increased after exposure to reversine but not AZ3146. PARP1 expression is altered only at the 8-cell stage in reversine-treated embryos (*** $P < 0.0004$, ** $P < 0.01$, * $P < 0.05$, t-test, error bars represent s.e.m. $n = 25$ embryos per treatment collected from 3 independent experiments).

To investigate how hypoxia exposure affects DNA damage and repair in mouse embryos, we again performed IF for γ H2A.X and PARP1 in 8-cell embryos and morula. We found that hypoxia exposure lowered the levels of γ H2A.X under all conditions, and minimized the differences between DMSO, reversine and AZ3146 treatments. In addition, PARP1 levels were also similar under all conditions, and relatively low in morula (**Fig. 3D-E**). Overall, these data suggest that hypoxia exposure lowers the accumulation of DNA damage under all conditions.

Hypoxia increases the proportion of euploid cells in the epiblast of mosaic blastocysts

Despite the high incidence of mosaicism in human pre-implantation embryos, the fate of aneuploid cells in mosaic embryos is incompletely understood. We previously used reversine to generate a mouse model of pre-implantation chromosome mosaicism and found that aneuploid (Reversine-treated) cells are eliminated in the EPI of mosaic embryos via apoptosis, starting from the mature blastocyst stage (Bolton et al., 2016). Whether this response reflects reversine treatment specifically, or aneuploidy more generally, is not known. Moreover, how hypoxia versus normoxia affects the outcomes is also not clear.

To address these questions, we created aggregation chimeras at the 8-cell stage that contained a 1:1 ratio of AZ3146-treated and control blastomeres (DMSO/AZ3146 chimeras), which is expected to reflect low-grade mosaicism, or of AZ3146-treated and reversine-treated blastomeres (reversine/AZ3146 chimeras), which is expected to reflect medium-grade mosaicism (**Fig. 4A**). We cultured these chimeric embryos in normoxia and hypoxia, and followed the fate of individual blastomeres by using transgenic mouse lines with the membrane markers mTmG (Muzumdar et al., 2007) and E-cadherin (Christodoulou et al., 2019). DMSO/AZ3146 and reversine/AZ3146 blastocysts displayed proper lineage allocation, embryo morphology and cavitation in both normoxia and hypoxia (**Fig. 4B-C**). In addition, DMSO/AZ3146 and reversine/AZ3146 embryos had comparable total cell numbers in their blastocysts, and blastocysts grown in hypoxia were again smaller than those grown in normoxia (**Fig. 4D-E**). Intriguingly, despite having fewer total cells, DMSO/AZ3146 blastocysts had more EPI cells when they were cultured in hypoxia compared to normoxia (**Fig. 4D-E**).

We quantified the proportion of AZ3146 cells in each lineage for each chimera. In DMSO/AZ3146-treated blastocysts, we found that 46.75% of the TE and 42.88% of the EPI originated from AZ3146-treated blastomeres, compared to only 28.57% of the PE (**Fig. 4F**). In reversine/AZ3146 chimeras, 63% of the TE and 78.4% of the EPI originated from AZ3146-treated cells, compared to only 40% of the PE (**Fig. 4F**). Thus, under normoxia, DMSO cells appear to outcompete AZ3146-treated cells, which in turn outcompete Reversine-treated cells in the TE and EPI. AZ3146-treated cells appeared to be at a competitive disadvantage in the PE in both contexts. Culturing DMSO/AZ3146 chimeras in hypoxia considerably increased the representation of AZ3146-treated blastomeres in the TE (49.06%) and PE (50%) but, strikingly, reduced their representation in the EPI (33.3%) (**Fig. 4G**). These altered frequencies reflect a higher number of DMSO-treated cells in the EPI and a lower number in the PE (**Fig. 4E**). Hypoxia exposure of reversine/AZ3146 embryos lowered the contribution of AZ3146-treated blastomeres to the TE (55.94%) and EPI (50%) but not the PE (46.43%) (**Fig. 4G**). We found that there was no correlation between the proportion of AZ3146-treated cells in the TE and the EPI in DMSO/AZ3146 and reversine/AZ3146 chimeras, under hypoxia or normoxia, and that hypoxia seemed to have a greater impact on the proportion of AZ3146 cells in reversine/AZ3146 chimeras (**Fig. S4**). Overall, these data suggest that hypoxia has lineage-specific effects on competitions between aneuploid and euploid cells and increases the contribution of euploid cells to the EPI.

Blastomeres in the 4-cell stage embryo display a lineage bias (Goolam et al., 2016). We considered that mosaicism generated before the 4-cell stage might influence lineage allocation. To test this possibility, we treated zygotes with reversine or AZ3146 during the first cleavage division.

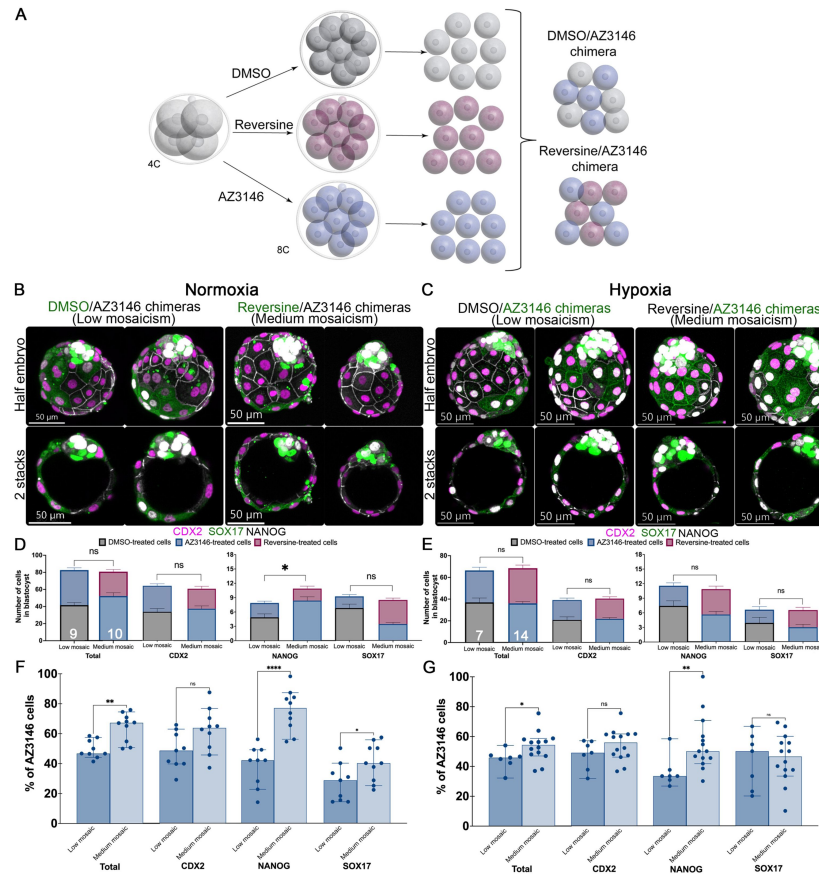


Figure 4

Hypoxia affects cell competition between diploid and aneuploid cells during pre-implantation development.

(A) Graphic representation of the cell competition experiments. Embryos were treated at the 4-cell stage with DMSO or Mps1 inhibitors reversine and AZ3146. After washing, 8-cell stage embryos were disaggregated, and re-aggregated to form chimeras containing a 1:1 ratio of DMSO/AZ314-treated blastomeres and Reversine/AZ314-treated blastomeres. Following aggregation, chimeras were cultured to the mature blastocyst stage (E4.5) and analyzed for lineage specification. For identification of the treatment, we use transgenic lines with membrane markers. Immunofluorescence for CDX2, NANOG, and SOX17 was performed to test lineage specification and allocation during (B) normoxia and (C) hypoxia. Lineage allocation quantification was based on the above markers. Importantly, (D) in normoxia, both chimeras have the same number of cells in all the lineages except the EPI. In addition, AZ3146-treated blastomeres outcompete reversine-treated blastomeres in medium-grade mosaics in the TE and the EPI. (E) Under hypoxia, both chimeras have similar number of cells in all the lineages. Yet, AZ3146-blastomeres do not outcompete reversine-treated blastomeres. Quantification of the contribution of AZ3146-treated blastomeres to the chimeras showed that, (F) under normoxia, compare with DMSO-treated cells, no preferential allocation of aneuploid cells occurs in the TE. In contrast, AZ3146-treated blastomeres increased their contribution when compare with reversine-treated blastomeres, but only significantly on the EPI and the PE. (G) During hypoxia, DMSO/AZ3146 chimeras did not change their behavior. Confirming a preferential allocation of diploid cells to the EPI. In Reversine/AZ3146 chimeras, AZ3146-treated blastomeres contribute similarly to reversine-blastomeres to the TE and PE but significantly increase contribution to the EPI. These results indicate that hypoxia favors the survival of reversine-induced aneuploid cells (**** $P < 0.00001$, ** $P < 0.01$, * $P < 0.05$, Mann-Whitney U-test, error bars represent s.e.m, $n > 7$ embryos per treatment collected from 3 independent experiments).

Importantly, reversine treatment at the zygote stage seems to strongly affect the morphology of the blastocysts (**Fig. S5A**). Consistent with this change in morphology, reversine-treatment reduced the number of cells in all lineages, particularly in the TE and PE (**Fig. S5B**). In contrast, AZ3146 treatment did not affect morphology or cell number in any of the lineages (**Fig. S5A-B**) ($n > 27$ embryos per treatment, cumulative of three independent experiments, **** $P < 0.0001$, * $P < 0.0044$, Mann–Whitney U-test). To evaluate how early generation of aneuploidies in the embryos affect cell competition, we generated DMSO/AZ3146, Reversine/DMSO, and Reversine/AZ3146/ aggregation chimeras at the 2-cell stage and cultured them under normoxia until the blastocyst stage (**Fig. S5C**). We found that all three 2-cell stage aggregation chimeras developed into blastocysts with proper lineage segregation and the presence of a cavity (**Fig. S5D**). Interestingly, the reversine-treated cells were extruded from the blastocyst in 54% of the reversine/DMSO chimeras and in 33% of the AZ3146/reversine chimeras, whereas no cell extrusion was observed in the DMSO/AZ3146 chimeras. Quantification of the proportion of AZ3146-treated cells in 2-cell stage derived chimeras (**Fig. S5E**) showed similar results as 8-cell stage derived chimeras (**Fig. 4F-G**). In DMSO/AZ3146-treated blastocysts, we found that 49.28% of the TE, 39% of the EPI and 33.33% of the PE originated from AZ3146-treated blastomeres (**Fig. S5E**). In reversine/AZ3146 chimeras, 68.24% of the TE, 68.83% of the EPI, and 66.67% of the PE originated from AZ3146-treated cells (**Fig. S5E**). In DMSO/reversine chimeras, 22.27% of the TE, 26.98% of the PE, and 31.7% of the EPI originated from reversine-treated cells. Overall, these results suggest that aneuploidy generated at different stages similarly affect the proportion of aneuploid cells in the blastocysts.

HIF1A inhibition reduces the frequency of aneuploid cells in DMSO/AZ3146 mosaic embryos

Our data suggest that hypoxia affects aneuploid-euploid cell competition in mosaic embryos (**Fig. 4**). To further assess the effect of HIF1A and, considering most human embryos are culture under hypoxic conditions, we decided to evaluate the effect of inhibiting HIF1A in mosaic embryos cultured in hypoxic conditions. We assembled DMSO/AZ3146 and Reversine/AZ3146 8-cell stage chimeras, treated them with the HIF1A inhibitor IDF-11774 (**Fig. 5A**, S6A-B), and then assessed lineage allocation in blastocysts. Treatment of DMSO/AZ3146 chimeras with IDF11774 does not affect lineage segregation in the blastocyst but reduces the cavity diameter (**Fig. 5B-C**). Additionally, diminish the contribution of AZ3146-treated cells to the TE, but DMSO-treated cells seem to compensate for this loss in the PE and EPI (**Fig. 5D**, **Fig. S6C**). Treating Reversine/AZ3146 chimeras with IDF11774 compromised blastocyst morphology and cavitation (**Fig. 5E-F**) and considerably increased the number of PE and EPI cells (**Fig. 5G**), without altering the frequency of AZ3146-treated cells (**Fig. S6D**). Taken together, our data suggest that HIF1A promotes the survival of AZ3146-treated cells in DMSO/AZ3146 chimeras and suggest that inhibiting HIF1A could increase the proportion of karyotypically normal cells in mosaic embryos.

Discussion

Aneuploidy is a common event during pre-implantation development (Allais & FitzHarris, 2022; Currie et al., 2022; Palmerola et al., 2022b). Here, we describe different strategies to generate aneuploidy in the early embryo. Our results suggest that: 1) the Mps1 inhibitor, AZ3146, generates aneuploid embryos with enhanced DNA repair and reduced stress responses when compared to reversine-treated embryos. 2) Lineage specific responses to aneuploidy are present during blastocyst development, as we showed previously (Bolton et al., 2016; Singla et al., 2020). 3) HIF1A contributes to the survival of TE and PE cells in AZ3146-treated embryos whereas PARP1 is particularly required in EPI cells in both reversine and AZ3146-treated embryos. 4) Hypoxia enhances DNA repair and affects aneuploid-euploid cell competition in mosaic embryos.

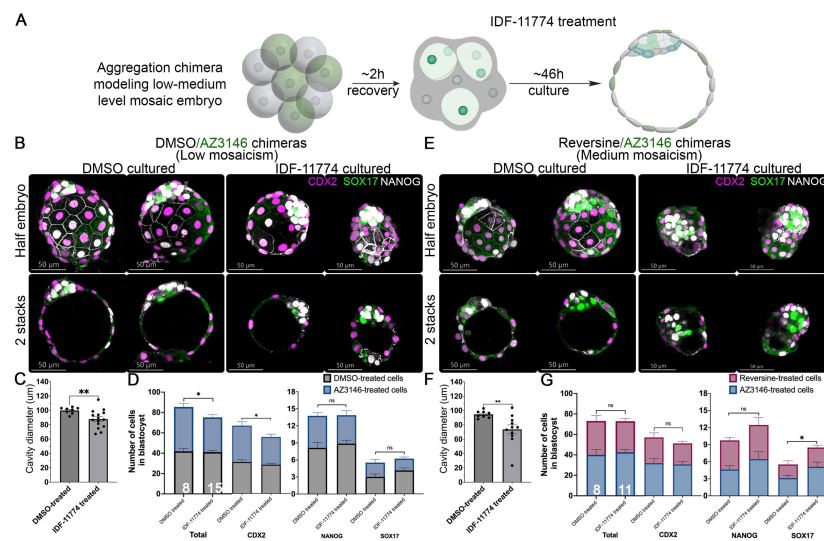


Figure 5

HIF1A inhibition increases the proportion of euploid cells in mosaic embryos.

(A) Schematic of HIF1A inhibition by IDF-11774 in 8-cell stage aggregation chimeras cultured in hypoxia. Immunofluorescence for CDX2, NANOG, and SOX17 was performed to test lineage specification and allocation. (B) HIF1A inhibition in low-grade mosaicism does not affect overall morphology but affects (C) cavitation. In contrast, (E) HIF1A inhibition in medium-grade mosaicism affects morphology, and (F) cavitation of the embryos. Lineage allocation quantification after IDF-1174 reveals a significant reduction of total cell number (D) in low-grade mosaicism as well as a reduction in cell number in the TE but not in the EPI and PE. In contrast, total cell number is not affected (G) in medium-grade mosaicism. However, IDF11-774 treatment appears to increase the cell number of the PE. These results show that HIF1A inhibition in hypoxic conditions differentially affect each type of mosaic embryos. (**P<0.005, *P<0.05, Mann-Whitney U-test, error bars represent s.e.m., n>8 embryos per treatment collected from 3 independent experiments).

Reversine-treated mouse embryos upregulate the mRNA levels of senescent markers p53 and p21 (Singla et al., 2020 [↗](#)) and fail to give rise to viable embryos (Bolton et al., 2016 [↗](#)). Our results further indicate that reversine downregulates PARP1, as previously observed in glioma (Hirakata et al., 2021 [↗](#)). Thus, reversine treatment may compromise embryo development due to a combined increase in DNA damage and decrease in DNA repair. In contrast, embryos treated with AZ3146 do not reduce PARP1 levels and have elevated HIF1A, which appears to improve survival of TE and PE cells and developmental potential.

DNA repair pathways are enhanced under hypoxic environments in somatic cells (Marti et al., 2021 [↗](#); Nakamura et al., 2022 [↗](#); Pietrzak et al., 2018 [↗](#)). Similarly, our data show that aneuploid embryos cultured in hypoxia had lower levels of DNA damage at the morula stage, in agreement with previous studies reporting lower levels of γH2A.X in mouse blastocysts recovered from the uterus and in blastocysts cultured in hypoxia compared to blastocysts cultured in normoxia (Houghton, 2021 [↗](#); Meuter et al., 2014 [↗](#)). Interestingly, recent studies suggest that hypoxia favors HIF1A's association with PARP1 in cancer cells, increasing DNA repair (Marti et al., 2021 [↗](#); Nakamura et al., 2022 [↗](#); Pietrzak et al., 2018 [↗](#)). In the future, a deeper understanding of the regulation of DNA repair in pre-implantation stages would allow us to understand the molecular response of embryos to aneuploid stress.

Confined placental mosaicism is commonly observed in human pregnancies, and refers to instances where the placenta is aneuploid but the fetus is euploid (McCoy, 2017 [↗](#)). We previously found that aneuploid TE cells have an increased cell cycle length and senescence, whereas aneuploid ICM cells have an increased frequency of apoptosis, in reversine-treated embryos (Singla et al., 2020 [↗](#)). Therefore, abnormal but viable TE cells could lead to confined placental mosaicism. Our results further suggest that aneuploid cells contributed mostly to the TE, followed by the EPI, are especially depleted from the PE in AZ3146-treated embryos under normoxia. Intriguingly, under hypoxia aneuploid cells were depleted from the EPI and increase in the PE. Overall, our data suggest that hypoxia may affect cell competition in a lineage-specific manner in the pre-implantation embryo, supporting an increased representation of euploid cells in the EPI relative to the PE and TE.

Finally, our results confirm that the proportion of aneuploid cells in the EPI and TE do not correlate, consistent with aneuploid survival in the TE but elimination from the ICM. These results are not surprising considering the low aneuploidy concordance rates between TE and ICM in human mosaic aneuploid embryos (30–40%) (Dahdouh & Garcia-Velasco, 2021 [↗](#)).

Since low- and medium- grade mosaic human embryos have similar developmental potential as fully euploid embryos (Capalbo et al., 2021 [↗](#); Greco et al., 2015 [↗](#)), understanding the molecular pathways involved in the responses to aneuploidy may have translational implications. Given that the extent and proportion of aneuploid cells in mosaic embryos impact survival (Capalbo et al., 2021 [↗](#)), we propose that a reduction in aneuploid cells in the blastocyst will favor a successful pregnancy. Here, we downregulated HIF1A as a strategy for reducing aneuploid cells number and found that treatment with IDF-11774 increased the proportion of euploid cells in mosaic embryos. In sum, our study reveals distinct environmental factors that affect the extent and proportion of aneuploid cells within mosaic embryos.

Materials and methods

Pre-implantation embryo culture

Mice were maintained according to national and international guidelines. Four- to six-week-old B6SJL/J female mice were injected with 7.5 IU PMSG followed by 7.5 IU hCG 48 h later, to induce superovulation. The females were then mated with B6CBAF1/J males or were indicated, with E-

cadherin GFP or mTmG transgenic males. Pre-implantation mouse embryos were recovered 24 h and 40 h after the hCG injection to obtain zygotes and 2-cell stage embryos, respectively in M2 medium (Sigma, M7167). We incubate the embryos in KSOM until 4-cell stage for the aneuploid treatment, around 53 h post hCG. When recovering zygotes, cumulus cells were removed with 0.3% hyaluronidase (Sigma, H4272) in M2. Embryos then were cultured in regular KSOM (Sigma, MR-106) at 37°C under 5% CO₂/air (Normoxia) or with pre-mixed 5% CO₂/ 5% O₂ balance with nitrogen, biologic atmosphere batch (Airgas #Z03NI9022000033)(Hypoxia). B6SJLF1/J females and B6CBAF1/J males were obtained from JAX laboratories. Females were received at 3–4 weeks of age and were maintained in the CALTECH animal facility, where they were housed with 5 same-sex littermates on a 12 h light/12 h dark cycle with food and water *ad libitum*. The temperature in the facility was controlled and maintained at 21 °C. All experimental procedures involving the use of live animals, or their tissues were performed in accordance with the NIH guidelines and approved by the Institutional Animal Care and Use Committee (IACUC) and the Institutional Biosafety Committee at the California Institute of Technology (Caltech, protocol number 1772). Reversine (Sigma, R3904), AZ3146 (Sigma, SML1427), PX-478 (Selleckchem, S7612), IDF-11774 (Selleckchem, S8771) and olaparib (Selleckchem, S1060) were dissolved in DMSO (Sigma, D2650) before use to specific concentration. They were respectively used at following final concentrations: 0.5 μM, 20 μM, 2 μM, 20 μM and 10 μM. Control embryos were incubated in the equivalent DMSO concentration. Drugs were dissolved in regular KSOM to the concentration of use. DMSO concentration in media should never pass 0.4%.

Immunofluorescence

Embryos were fixed in 4% PFA (Thermo Scientific, AA47340) for 20 min at room temperature (RT). Following by three washes with 0.1% Tween-20 (Sigma, P1379) dissolved in PBS (PBST). The embryos were then permeabilized with 0.3%, Triton X-100 (Sigma, X100) in PBS. Washes were then performed three times in PBST before embryos were transferred to blocking solution (3% BSA in PBS) for at least 3h at RT. Incubation with primary antibodies was performed in blocking solution overnight at 4°C. Next day, washes were performed three times in PBST before incubation with Alexa Fluor secondary antibodies (Thermo Fisher Scientific, 1:500) in blocking solution for 2 h. Washes were performed three times before incubation with DAPI (Thermo Fisher Scientific) for 5 min. Washes after DAPI were performed two times in PBST before the final incubation in M2. Embryos were then mounted in M2 micro-drops on 35 mm glass bottom dish (MATTEK, P35G-1.5-14-C). Confocal imaging was carried out using Leica SP8, 40x objective, 1 μM Z-step. Image files were viewed and analyzed using ImageJ and IMARIS 9.9 software.

Primary antibodies used: mouse anti-CDX2 (Biogenex, 1:500), rabbit anti-NANOG (1:500; Abcam), goat anti-SOX17 (1:300; R&D Systems), rabbit anti-HIF1A (1:300, Novusbio), mouse anti-PARP1 (1:500, Proteintech) and rabbit anti- Phospho- Histone H2AX (1:500, R&D Systems).

In situ chromosome counting

For determination of aneuploidy, treated 8-cell stage embryos were synchronized in metaphase by 10h treatment with 0.03 μg/mL colcemid (Cayman, 15364) diluted in KSOM. Following by 1h treatment with 10 μM Mg132 (Selleckchem, S2619) in KSOM. Finally, 1 h treatment with 5 μM dimethylenastron (MedChemExpress, HY-19944) and 10 μM Mg132 in KSOM. Synchronized embryos were then fix in 2% PFA for 20 min, permeabilized with PBST for 15 min and blocked 3h at RT before incubation with human anti-centromere protein antibody (1:300, Antibodiesinc 15-234) overnight at 4°C. Next day, three washes with PBST were performed before 2 h incubation with goat anti-Human secondary antibody, Alexa Fluor 647 (1:400, Invitrogen A-21445) in blocking solution. Washes were performed three times before 20 min incubation with DAPI (1:500) and Alexa Fluor 488 Phalloidin (1:300, Invitrogen A12379). Washes then were performed three times before clearing overnight with AF1 plus (Citifluor, AF1/DAPI-15). Embryos were then mounted in

AF1 plus micro-drops on 35 mm glass bottom. Confocal imaging was carried out using Leica SP8, 60x objective, 0.5 μ M Z-step. Image files were viewed and analyzed using ImageJ and IMARIS 9.9 software.

Embryo transfers and post implantation recovery and biopsy

Embryo transfer were performed as described previously (Bermejo-Alvarez et al., 2014 [DOI](#)). CD1 females and vasectomized CD1 males were obtained from Charles River. Reversine and AZ3146-treated blastocyst were transfer into 2.5-days pseudopregnant CD1 females. Around 16 AZ3146-treated embryos were transfer in the right uterine horn, whereas around 16 reversine-treated embryos were transfer in the left uterine horn of the same female as control. To evaluate implantation and embryo survival potency, uterine horns were recovered 5 days after surgery. Deciduas were dissected and post implantation embryos at stage E9.5 were recovered into PBST on ice. Embryos were fixed in 4% PFA at RT for 1 h, followed by washes through PBST before imaging. Images were taken using an Olympus LS stereo microscope with a 10x objective. Uterine transfers were performed in accordance with the NIH guidelines and approved by the Institutional Animal Care and Use Committee (IACUC) and the Institutional Biosafety Committee at the California Institute of Technology (CALTECH).

qRT-PCR

Around 14 to 17 morulas and blastocysts were collected for quantitative reverse transcriptase polymerase chain reaction (qRT-PCR). Total RNA from morulas was obtained using the NucleoSpin RNA Plus XS kit (Takara, 740990.10). Whereas total RNA from blastocyst was extracted using the Arcturus PicoPure RNA Isolation Kit (Thermo Fisher, KIT0204). qRT-PCR was performed using the Power SYBR Green RNA-to-CT 1-Step Kit (Applied Biosystems, 4389986) in a StepOne Plus Real-time PCR machine (Applied Biosystems). The following program was used: 30 min 48 °C (reverse-transcription) followed by 10 min 95 °C followed by 45 cycles of 15 s 95 °C (denaturing) and 1 min 60 °C (annealing and extension). The ddCT method was used to determine relative levels of mRNA expression, with Ppia as an endogenous control (Gu et al., 2014 [DOI](#)). Primers were obtained from IDT.

Name	Sequence
p53_Fw	GTCACAGCACATGACGGAGG
p53_Rv	TCTTCCAGATGCTCGGGATAC
Hif1a_Fw	CCTGCACTGAATCAAGAGGTTGC
Hif1a_Rv	CCATCAGAAGGACTTGCTGGCT
Ppia_Fw	GAGCTCTGAGCACTGGAGAGA
Ppia_Rv	CCACCCTGGCACATGAAT

Generation of chimeric embryos

Chimeric embryos were generated following a previous publish protocol (Eakin & Hadjantonakis, 2006 [DOI](#)). Briefly, wild-type, E-cadherin GFP or mTmG 8-cell stage embryos were transfer to micro-drops of M2 after aneuploid treatment. Zona pellucida was removed by treatment with acidic Tyrode's solution (Sigma, T1788). The embryos were then incubated in Ca²⁺/Mg²⁺-free M2 (made in house) for 5 min and then disaggregated into individual blastomeres by gentle mouth pipetting.

Mosaic chimeras were formed by aggregation of four and four (1:1) of each treatment. Low mosaicism DMSO/AZ3146-treated blastomeres (1:1). Whereas medium mosaicism Reversine/AZ3146-treated blastomeres (1:1). Culture of the chimeras were performed under normoxic and hypoxia condition in KSOM for 48 h to reach blastocyst stage.

Statistical analysis

The statistical tests used are indicated in the corresponding figure legends. Calculations were carried out in Microsoft Excel and data analysis and visualization in Prism 9 software. All graphs show mean values, error bars: s.e.m.

Supplementary figures

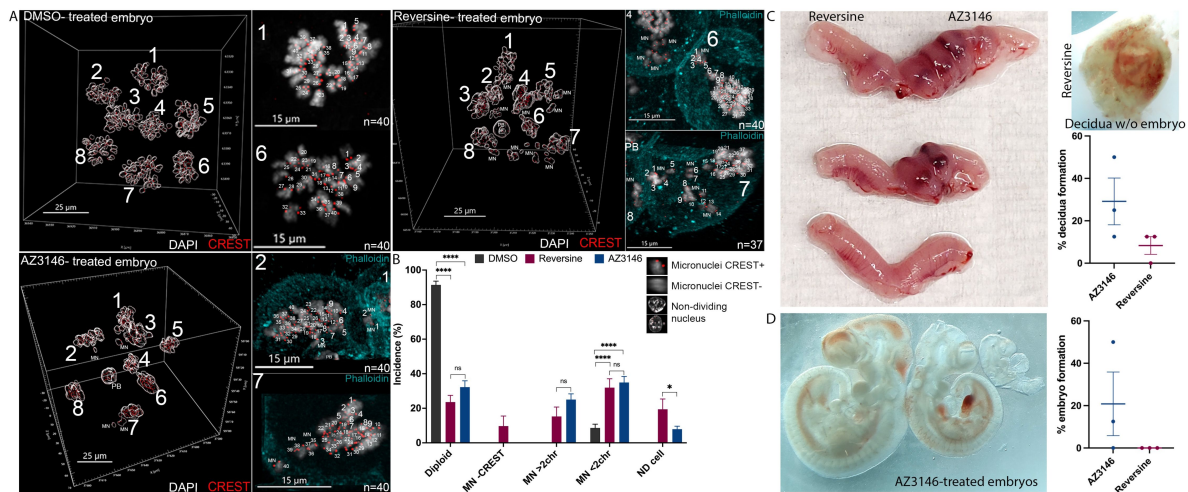


Figure S1

AZ3146-treated embryos are aneuploid and can develop to post-implantation stages.

(A) *In situ* chromosome counting of 8-cell stage embryos after drug treatments. Representative images of individual embryos in the different treatments. DNA (white) is visualized by DAPI staining and kinetochores (red) by CREST antibody. PB (polar body), MN (micronuclei). Each chromosome presents 2 CREST dots. The large number is the cell ID and the smaller number indicates the number of chromosomes, "n". Diploid blastomeres present 40 pairs of CREST dots. Phalloidin (cyan) was used for cell segmentation. (B) Quantification of the incidence of aneuploid events, designated as those with more than 2 chromosomes (MN>2chr), less than 2 chromosomes (MN<2chr), without kinetochores (MN-CREST) and non-dividing nucleus (ND cell). We observed an increase in MN following reversine and AZ3146 treatment. Importantly, reversine-treated cells have a higher number of ND cells and MN-CREST staining. (n=256 cell DMSO treatment, n=304 cells AZ3146 treatment, n=144 cells reversine treatment, ****P<0.00001, *P<0.05, Mann-Whitney U-test, error bars represent s.e.m.). (C) Embryo transfer experiments show higher implantation events (% decidual formation) in AZ3146-treated embryos. The left uterine horn was used for transfer of reversine-treated blastocyst and the right uterine horn for AZ3146-treated blastocyst. (n=7 embryos transplanted per uterine horn, repeated experiments two times) (D) Only AZ3146-treated blastocyst can develop post-gastrulation E9.5 fetuses (% embryo formation).

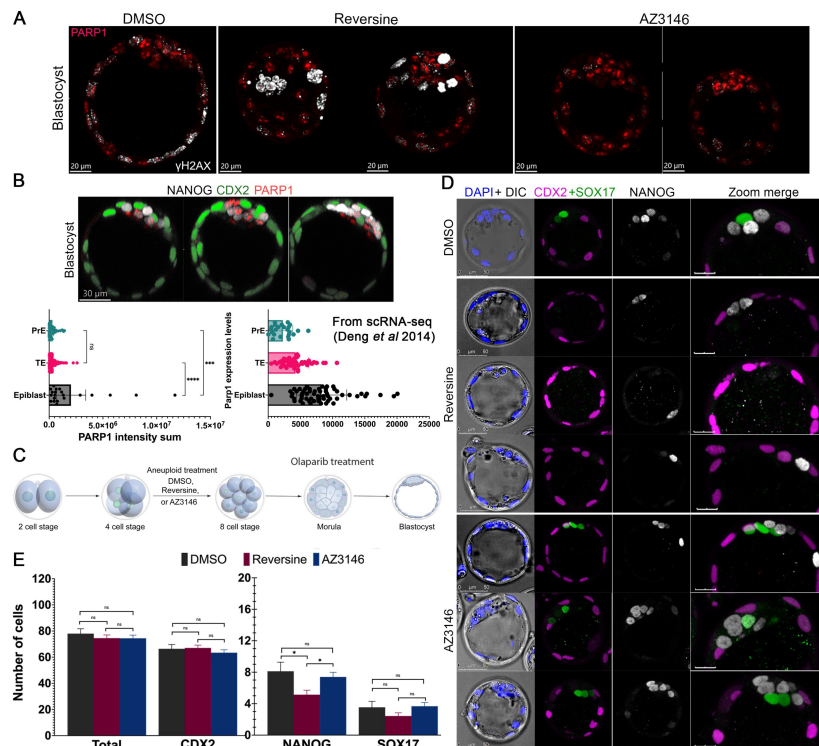


Figure S2

PARP1 is required for ICM development.

(A) Immunofluorescence against PARP1 (red) and γ H2A.X (white) in blastocyst after drug treatments. Notice accumulation of DNA damage in reversine-treated blastocyst. (B) Immunofluorescence against PARP1 (red), NANOG (white, epiblast) and CDX2 (green, trophectoderm). Analysis of PARP1 intensity shows an increase in the EPI lineage (**** $P < 0.00001$, *** $P < 0.0006$, Mann-Whitney U-test, error bars represent s.e.m, $n > 21$ embryos per treatment collected from 3 independent experiments). To confirm these observation, scRNA-seq (Deng et. al 2014) shows enrichment of *Parp1* mRNA specifically in the EPI. (C) Graphic representation of 4-cell embryos treated with DMSO and aneuploid drugs. Downregulation of PARP1 was achieved by treatment with Olaparib from morula to blastocyst stage. (D) Effects of the inhibition of PARP1 function were assessed by immunofluorescence against CDX2 (TE), NANOG (EPI) and SOX17 (PE) in the different treatments. Importantly, Olaparib treatment does not affect blastocyst morphology. (E) Lineage analysis at blastocyst stage shows a specific effect on the survival of epiblast and PE cells in reversine-treated embryos. Ratio of cells in each lineage in the blastocyst is normalized based on DMSO treatment. (* $P < 0.05$, Mann-Whitney U-test, error bars represent s.e.m $n = 15$ embryos per treatment collected from 3 independent experiments).

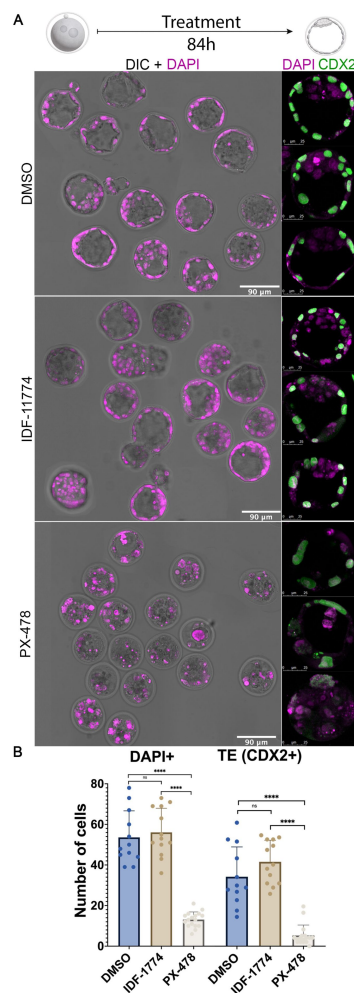


Figure S3

Pharmacological inhibitors of HIF1A have distinct effects on mouse pre-implantation embryos.

(A) Inhibition of HIF1A from zygotes to blastocyst stage with PX-478 or IDF-11774. Immunofluorescence against CDX2 (TE) and DAPI (magenta) was used to assess blastocyst development. Importantly, IDF-11774 does not compromise pre-implantation development, whereas PX-478 compromised the survival and overall morphology of the blastocyst. (B) Analysis of the number of trophoblast cells (CDX2) and total number (DAPI) shows a deleterious effect of PX-478 during blastocyst development (**** $P < 0.0001$, Mann-Whitney U-test, error bars represent s.e.m, $n > 12$ embryos per treatment collected from 3 independent experiments).

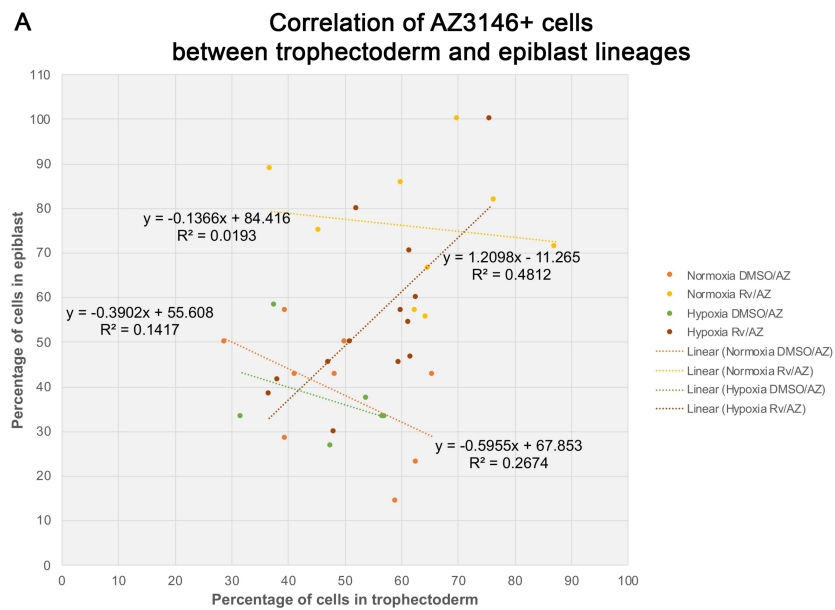


Figure S4

Correlation plot of AZ3146-treated cells in the epiblast and trophectoderm.

Scatter plot of AZ3146-treated cells in epiblast and trophectoderm in DMSO/AZ3146 and Reversine/AZ3146 mosaic blastocysts cultured under normoxic and hypoxic conditions.

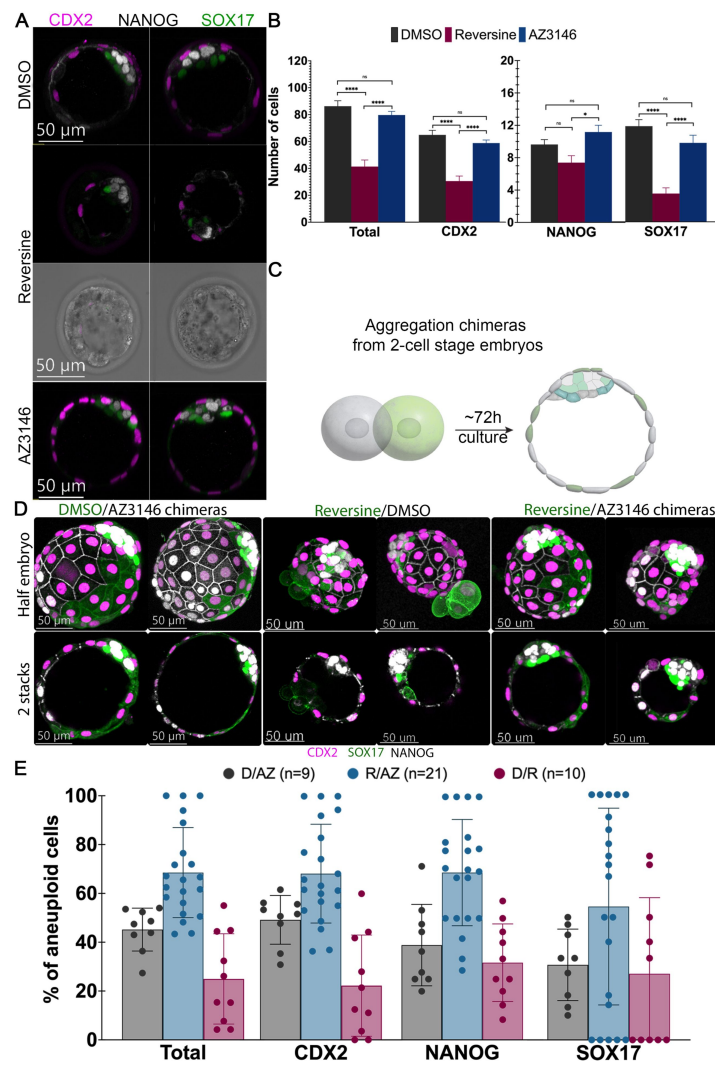


Figure S5

Aneuploidies generated before the 2-cell stage does not affect lineage specification in mosaic embryos.

(A) Immunofluorescence for CDX2, NANOG, and SOX17 was performed in blastocysts after zygote treatment with Mps1 inhibitors. Importantly, reversine treatment at zygote stages affects morphology of blastocyst. (B) Lineage analysis of blastocyst shows no effect in AZ3146-treated embryos. Whereas reversine treatment reduces the cell number to almost half. Ratio of cells in each lineage in the blastocyst is normalized based on DMSO treatments. ($n > 27$ embryos per treatment collected from 3 independent experiments, **** $P < 0.0001$, * $P < 0.0044$, Mann-Whitney U-test, error bars represent s.e.m). (C) Aggregation chimeras at 2-cell stage were created by using transgenics lines with membrane markers: mTmG (green) and E-cadherin (white). Subsequently, (D) immunofluorescence for CDX2, NANOG, and SOX17 was performed to quantify lineage allocation. Our results show no effect on embryo morphology in our chimeras. However, reversine/DMSO (D/R) chimeras seems to have increase events of cell extrusion and reversine/AZ3146 chimeras are smaller in number of cells compare with DMSO/AZ3146 chimeras. (E) Data show the proportion of AZ3146-treated cells in DMSO/AZ3146 mosaic blastocysts (grey) and in reversine/AZ3146 mosaic blastocysts (blue), as well as the proportion of reversine-treated cells in DMSO/reversine mosaic blastocysts (red).

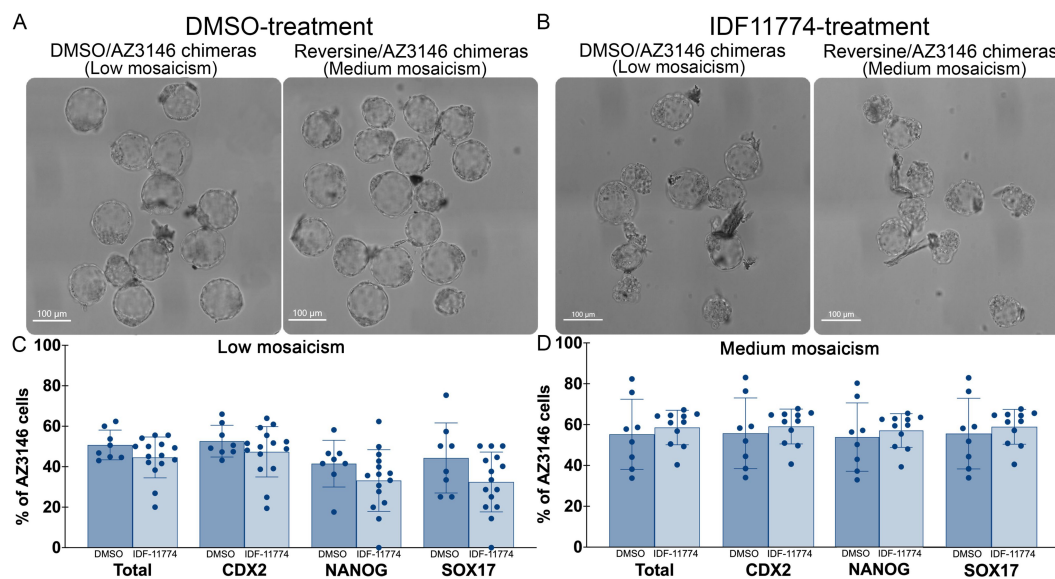


Figure S6

IDF-11774-mediated inhibition of HIF1A increases the proportion of euploid cells in mosaic embryos.

Brightfield images of low- and medium-mosaic chimeras grown in (A) DMSO and (B) IDF-11774. Quantification of the contribution of AZ3146-treated blastomeres to the chimeras showed that, (C) in DMSO/AZ3146 mosaics, IDF-11774 treatment reduces the proportion of AZ3146-treated blastomeres, whereas (D) in reversine/AZ3146 mosaics, IDF-11774 treatment slightly increases the proportion of AZ3146-treated blastomeres. These results may indicate that HIF1A inhibition impairs the less fit population in chimeric embryos.

Acknowledgements

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Additional information

Author contributions

E.S.V designed and conducted the experiments; analyzed and interpreted the data. E.S.V and M.Z.G conceived the project. M.E.B and M.Z.G supervised the study.

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Reviewer #1 (Public review):

Summary:

This paper developed a model of chromosome mosaicism by using a new aneuploidy-inducing drug (AZ3146), and compared this to their previous work where they used reversine, to demonstrate the fate of aneuploid cells during murine preimplantation embryo development. They found that AZ3146 acts similarly to reversine in inducing aneuploidy in embryos, but interestingly showed that the developmental potential of embryos is higher in AZ3146-treated vs. reversine-treated embryos. This difference was associated with changes in HIF1A, p53 gene regulation, DNA damage, and fate of euploid and aneuploid cells when embryos were cultured in a hypoxic environment.

Strengths:

In the current study, the authors investigate the fate of aneuploid cells in the preimplantation murine embryo using a specific aneuploidy-inducing compound to generate embryos that were chimeras of euploid and aneuploid cells. The strength of the work is that they investigate the developmental potential and changes in gene expression profiles under normoxic and hypoxic culture conditions. Further, they also assessed how levels of DNA damage and DNA repair are altered in these culture conditions. They also assessed the allocation of aneuploid cells to the divergent cell lineages of the blastocyst stage embryo.

Weaknesses:

The authors have still not addressed the inconsistent/missing description for sample size, the appropriate number of * for each figure panel, and the statistical tests used.

The authors assign 5% oxygen as hypoxia. This is not the case as the in vivo environment is close to this value. 5% is normoxia. Clinical IVF/embryo culture occurs at 5% O₂. Please adjust your narrative around this.

<https://doi.org/10.7554/eLife.101912.2.sa2>

Reviewer #2 (Public review):**Summary:**

This study by Sanchez-Vasquez is a very innovative approach to induce aneuploidy and then study the contribution of treated cells to different lineages, including post implantation. It connects well to the authors previous work to induce mosaic aneuploidies. The authors identify sensitivity to HIF1a loss in treated embryos with likely aneuploidy. This work is part of an important line of work with evaluates the consequences of aneuploidy in mammalian embryo.

Weaknesses:

Given that this is a study on the induction of aneuploidy, it would be meaningful to assess aneuploidy immediately after induction, and then again before implantation. This is also applicable to the competition experiments on page 7/8. What is shown is the competitiveness of treated cells. Because the publication centers around aneuploidy, inclusion of such data in the main figure at all relevant points would strengthen it. There is some evaluation of karyotypes only in the supplemental - why? Would be good not to rely on a single assay that the authors appear to not give much importance.

<https://doi.org/10.7554/eLife.101912.2.sa1>

Author response:

The following is the authors' response to the original reviews.

We deeply appreciate the reviewer comments on our manuscript. We have proceeded with all the minor changes mentioned. We also want to emphasize three major points:

(1) Reversine has been shown to have several off-targets effects. Including inducing apoptosis (Chen et al. J Bone Oncol. 2024).

(2) Hypoxia varies from 2% to 6%. Our definition of hypoxia is 5% concentration of oxygen with 5% concentration of CO₂, taking into consideration the standard levels of oxygen in the IVF clinics. Physiological oxygen in mouse varies from ~1.5% to 8%.

(3) Natale et al. 2004 (Dev Bio) and Sozen et al. 2015 (Mech of Dev) described that inhibition of p38 deeply affect the development of pre-implantation embryos after the 8-cell stage. For this reason, comprehensible dissect the interaction between p53, HIF1A and p38 during aneuploid stress is challenging. We do not discard a double function of p38 during lineage specification and in response to DNA damage.

Reviewer #2 (Recommendations for the authors):

(1) Line 69: Please add the species used in your cited publications (murine).

Fixed

(2) Line 72: Consider changing "Because" to "As".

Fixed

(3) Line 88: "from the nuclei" - please refer to where the reader may find the example provided (Figure S1A).

Fixed

(4) Line 89: This should be Figure S1B as no quantification is presented in S1A. S1A only contains examples of micronuclei.

Fixed

(5) Line 91: Refer to Figure S1A.

Fixed

(6) Line 91-93: Are these numbers correct? The query arises from the numbers presented in Figure S1B. Please define how the median was calculated; is it micronuclei CREST+ plus micronuclei CREST-?

Fixed. We did not differentiate in these percentage the presence of CREST.

(7) Line 95: extra/missing bracket?

Fixed

(8) Line 88-91:

[a] Regarding the number of cells with micronuclei in this text, please clarify your sample size and how the percentages were calculated as they currently do not align (e.g., are these the total number of embryos from a single experimental replicate?).

Also, different numbers are found here and in the figure legend: (DMSO-22/256 cells from 32 embryos; Rev-82/144 cells from 18 embryos; AZ-182/304 cells from 38 embryos) vs. Fig S1 legend (DMSO-n=128 cells; Rev-72 cells; AZ-152 cells).

[c] Is the median calculated using the numbers presented above? If yes, then the numbers do not tally, please check (DMSO-22/256 cells=8.6%; Rev-82/144 cells=56.9%; AZ-182/304 cells =59.9%) vs. Line 91-93: DMSO=12.5%, Rev=75%; AZ=62.5% blastomeres had micronuclei.

The percentage represents the average of aneuploidy per embryo after normalization.

See table for DMSO. This number represents the average of aneuploid cells each aneuploid embryo has. Notice that some embryos are fully diploid. Some have more that 12.5% -> 25%. Most of the aneuploid embryos have 12.5% of aneuploidy. It is not black and white as how many aneuploid cell there is in the sample but a full understanding of how aneuploid are the aneuploid embryos in each sample.

Author response image 1.

Treatment	Position	Diploid	MN - count	MN - %	MN - 2	Non dividing	Total		Treatment	Position	Diploid	MN - count	MN - %	MN - 2	Non dividing	Total	NO PLOID CELL	ANEUPLOID
DMSO	1	8	0	0	0	0	8		DMSO	1	100	0	0	0	0	100	8.75	12.5
	2	8	0	0	0	0	8			2	100	0	0	0	0	100		
	3	8	0	0	0	0	8			3	100	0	0	0	0	100	MN - count	0
	4	7	0	0	1	0	8			4	87.5	0	0	12.5	0	100	MN - %	0
	5	7	0	0	1	0	8			5	87.5	0	0	12.5	0	100	MN - 2	8.58375
	18	8	0	0	0	0	8			18	100	0	0	0	0	100	Non dividing	0
	7	8	0	0	0	0	8			7	100	0	0	0	0	100		
	8	7	0	0	1	0	8			8	87.5	0	0	12.5	0	100		
	9	7	0	0	1	0	8			9	87.5	0	0	12.5	0	100		
	10	7	0	0	1	0	8			10	87.5	0	0	12.5	0	100		
	17	8	0	0	0	0	8			17	100	0	0	0	0	100		
	12	7	0	0	1	0	8			12	87.5	0	0	12.5	0	100		
	13	8	0	0	0	0	8			13	100	0	0	0	0	100		
	14	6	0	0	2	0	8			14	75	0	0	25	0	100		
DMSO	15	6	0	0	2	0	8		DMSO	15	75	0	0	25	0	100		
	16	7	0	0	1	0	8			16	87.5	0	0	12.5	0	100		
	11	7	0	0	1	0	8			11	87.5	0	0	12.5	0	100		
	6	8	0	0	0	0	8			6	100	0	0	0	0	100		
	19	7	0	0	1	0	8			19	87.5	0	0	12.5	0	100		
	20	7	0	0	1	0	8			20	87.5	0	0	12.5	0	100		
	21	7	0	0	1	0	8			21	87.5	0	0	12.5	0	100		
	22	7	0	0	1	0	8			22	87.5	0	0	12.5	0	100		
	23	6	0	0	2	0	8			23	75	0	0	25	0	100		
	28	6	0	0	2	0	8			28	75	0	0	25	0	100		
	25	7	0	0	1	0	8			25	87.5	0	0	12.5	0	100		
	26	8	0	0	0	0	8			26	100	0	0	0	0	100		
	27	7	0	0	1	0	8			27	87.5	0	0	12.5	0	100		
	38	8	0	0	0	0	8			38	100	0	0	0	0	100		
	39	8	0	0	0	0	8			39	100	0	0	0	0	100		
	30	8	0	0	0	0	8			30	100	0	0	0	0	100		
	31	8	0	0	0	0	8			31	100	0	0	0	0	100		
	32	8	0	0	0	0	8	256		32	100	0	0	0	0	100		

(9) Line 108:

[a] "n=28 per treatment" please clarify whether this refers to the number of embryos or cells and also add how many independent replicate experiments this data is representative of. as the text only refers to Figure 1C you can remove the P-values for ** and *.

Number of embryos. Fixed

(10) Line 111: Suggest citing Figure 1C at the end of the sentence.

Fixed

(11) Line 118-119: the reference to figures require updating to ensure they refer to the appropriate figure; ...decidua (Figure S1C)...viable E9.5 embryos (Figure S1D).

Fixed

(12) Line 126: A description of the data in Figures 1D and 1E is missing. Also, consider describing the DNA damage observed in the DMSO control group. Visually, it appears that DNA damage reduces from the 8-cell to the morula stage (Figure 1E) but increases at the blastocyst stage (Figure S2A)? Point for discussion for a normal rate of DNA damage?

Agree, there is some DNA damage in the TE in blastocyst

| (13) Line 134: 8 EPI and 4 PE cells in what group?

Fixed: DMSO-treated embryos

| (14) Line 137: Could this also suggest that AZ and reversine induce DNA damage through a different mechanism/pathway, resulting in the differential impact observed? Despite both being inhibitors of Mps1.

This is a possibility.

| (15) Line 153: the legend for Figure 2A says the Welch t-test was performed, but the Mann-Whitney U-test was stated here. Which is correct?

Welch's t-test

| (16) Line 155: ...at the blastocyst stage. Compared to what?

DMSO-treated embryos

| (17) Line 160: Data in Figure 2B requires the definition of P-values for *, **, ****. Please add one for **** and remove the one for ***.

Fixed

| (18) Line 173-174: Data in Fig. 4 requires the definition of the P-values for ****. Please remove the others.

Fixed

| (19) Line 180: Instead of jumping across figures, this section would benefit from stating the numbers directly to allow for an accurate comparison, e.g. 64 and 7 in Figure 2D vs. X and Y in Figure 1C.

| (20) Line 187: *Hif1a* should be italicised.

Fixed

| (21) Line 197: Based on the description here, I believe you are missing a reference to Figure 1A.

Fixed

| (22) Line 203: Instead of jumping across figures, this section would benefit from stating the numbers directly to allow for accurate comparison, "particularly in the TE and PE" (67 vs 54; and 11 vs 6, respectively).

| (23) Line 209-210:

| [a] "...lowered the number of γ H2AX foci..." is this a visual observation as quantification was performed for γ H2AX intensity, not quantification of foci?

| A description for PARP1 levels in morula stage embryos was presented ("...relatively low in morula), but not for γ H2AX levels at this stage of development. Missing description?

Fixed

(24) Line 235: *This sentence would benefit from being specific about the environmental conditions...eg "Under normoxia, DMSO/AZ3146-treated..."*,

(25) Line 238: *The sentence should reference Figure 4F not 4G.*

Fixed

(26) Line 242-243:

[a] "slightly increased... in the TE (49.06%) and PE (50%) but, strikingly, reduced... EPI (33.3%)" compared to what and in which figure?

Assuming you are comparing normoxia (4F) to hypoxia (4G), the numbers change for the TE (46.75% to 49.06%, respectively), EPI (42.88% to 33.3%, respectively), and PE (28.57% to 50%, respectively); yet these data were described as "strikingly different" for EPI (9.58 decrease) but only "slightly increased" for PE (21.42 increase). Suggest using appropriate adjectives to describe the results.

Fixed

(27) Line 256: *It is stated in line 255 that treatment was performed at the zygote stage, yet this sentence says reversine treatment occurred at the 2-cell stage? Which is correct? Please amend appropriately. Refer to the comment below regarding adding a schematic to aid readers*

Fixed

(28) Line 259: *"n>27 per treatment" please clarify whether this refers to the number of embryos or cells and also add how many independent replicate experiments this data is representative of. Data in Figures S5A-B requires a definition of P-values for *, ***. Please remove for **, ***.*

Fixed

(29) Line 261: *AZ3146/reversine stated here, the figure shows Reversine/AZ3146. Please consider being consistent.*

Fixed

(30) Line 263: *"... normal morphology and cavitation (Figure S5D); however the image presented for Rev/DMSO and Rev/AZ3146 chimeras appear smaller with a distorted/weird shape when compared to DMSO/AZ. I believe the description does not match the images presented.*

Fixed

(31) Line 267: *"...similar results as 8-cell stage derived chimeras"; however, there is only a reference to Fig S5E which depicts 2-cell/zygote stage (see comment above for line 256 regarding required clarification of stage of treatment) derived chimeras. There is also a missing reference to Figure 4B, D, and/or F?*

Fixed

| (32) Line 271: add a reference to Figure S5E.

Fixed

| (33) Line 283: "AZ3146/reversine" should be "Reversine/AZ3146" to match the figure.

Fixed

| (34) Line 284: Figures 5E-F show both morphology and cavitation; the text should reflect this.

Fixed

| (35) Line 281-285: I think this text requires editing to improve clarity. It is difficult for this reader to understand the authors' interpretation of the results....inhibiting HIF1A reduces morphology and cavitation. That's correct. However, this also diminished the contribution of AZ3146-treated cells to all 3 cell lineages; this is not quite accurate. AZ3146-treated cells were significantly reduced in total cell numbers because TE was significantly reduced. It is not appropriate to generalise this result to all 3 lineages, as EPI and TE appear to increase AZ's contribution following IDF treatment, albeit non-statistically significant.

Fixed

| (36) Line 320: citation?reversine-treated embryos. Is this referring to your previous publication...Bolton 2016?

Fixed

| (37) Line 344: missing space between 7.5 and IU.

Fixed

| (38) Line 358: animal ethics approval number/code missing.

Fixed

| (39) Line 397: missing space between "...previously" and "(Bermejo...".

Fixed

| (40) Line 417: missing space between "...control" and "(Gu et...".

Fixed

| (41) Line 421: missing space between "protocol" and "(Eakin...".

Fixed

| (42) Line 427-429: Medium-grade mosaic chimeras were referred to as DMSO:AZ:Rev (3:3:2) here; but Figure 4 and associated legend says otherwise. Please amend appropriately. Were all medium mosaics generated in this manner? As I could only find

Rev/AZ chimeras; my understanding of the Rev/AZ chimeras is 1:1 Rev:AZ instead of 3:2:3 DMSO:Rev:AZ.

Fixed

(43) Line 428: "reversine-treaded: please correct spelling.

Fixed

(44) Line 593: "n=28 per treatment" Please clarify whether this refers to the number of embryos or cells and also add how many independent replicate experiments this data is representative of.

Fixed

(45) Line 597: "through morula stage" when compared to what group?

DMSO-treated embryos

*(46) Line 598: Data in Figure S5A-B requires the definition of P-values for **, ***, ****. Please remove for *. Please define the error bars. SEM/95% confidence interval?*

Fixed

(47) Line 604-607: Regarding 2B, no statistical test is stated yet Mann-Whitney was stated in Line 160 of the results section. Please confirm which test was used and include it in both sections for consistency.

Fixed

(48) Line 608: "Chemical downregulation of HIF1A"... this is not described in the results/methods section or shown in the figure. Please amend all sections for accuracy.

Fixed

(49) Line 613: please change "effect in" to "effect on".

Fixed

*(50) Line 614: Please clarify the number of embryos or cells and also add how many independent replicate experiments this data is representative of. Data in Figure 2 also requires a definition of P-value for ****.*

Fixed

*(51) Line 625: Please clarify the number of embryos or cells and also add how many independent replicate experiments this data is representative of. Data in Figure 3 also requires a definition of P-value for ****.*

Fixed

(52) Line 627: description requires editing to improve accuracy "...is only slightly increased at the 8-cell stage after exposure to reversine and AZ3146". However, the

results show significantly higher DNA damage with Reversine treatment, but not with AZ when compared to DMSO. Please amend accordingly.

Fixed

(53) Line 629: Please define the error bars. SEM/95% confidence interval?

Fixed

(54) Line 634-635: it is written here that chimeras were made from 1:1 DMSO/AZ3146 and Reversine/DMSO; but Figure 4A shows 1:1 DMSO(grey):AZ3146(blue), and Reversine(red):AZ3146(blue), which contradicts the legend + method section; see comments for Line 427-429. Please amend these sections accordingly.

Fixed

(55) Line 648: reversine/AZ3146 chimeras? Refer to comments above.

Fixed

(56) Line 649-650: ...AZ-treated blastomeres contribute similarly to reversine-blastomeres to the TE and EPI but significantly increase contribution to the EPI? Please add the appropriate comparison group.

Fixed

(57) Line 652: Please clarify the number of embryos or cells and also add how many independent replicate experiments this data is representative of.

Fixed

(58) Line 664: Please clarify the number of embryos or cells and also add how many independent replicate experiments this data is representative of.

Fixed

*(59) Line 675-677: FigS1B legend requires a definition of P-value for * and ****, can omit ***

Fixed

(60) Line 678-680: FigS1C and S1D legend: sample size and replicates? Only mentioned in Lines 117-120, which requires back calculation.

Fixed

*(61) Line 682-694: (1) Fig. S2B legend: missing P-value description for *** and ****; statistical test not stated, please add. Also, Figure S2E, only requires the definition for *, and can omit others.*

Fixed

*(62) Line 702: FigS3B: missing description for ****, omit others.*

Fixed

(63) Line 704-705: missing description for Rev/AZ group and hypoxia vs. normoxia conditions.

Fixed

(64) Line 712-713: "n>27 per treatment" Please clarify whether this refers to the number of embryos or cells and also add how many independent replicate experiments this data is representative of. Data in Figure S5 requires the definition of P-values for *, ****. Please remove for **, ***.

Fixed

(65) Line 713-715: could benefit from a description of which were marked from mTmG; e.g. why is DMSO, Rev, Rev in Green for [D]; does this mean 2-cell stage chimeras were only made with embryos treated with DMSO and Reversine? Has it been tested if you did this with AZ3146, do the proportions remain the same? This would be interesting to know.

DMSO and reversine are in green because they are the cells mark with green in the chimeras. We also did chimeras with AZ3146. Hope this clarifies.

(66) Line 719-721: why is there a difference between the proportion of aneuploid cells for the different chimeras? AZ in D/AZ, and R/AZ groups; while only R in D/R group? Is this because you only count those that were marked with mTmG (e.g. based on [Fig S5D])?
(67) Line 724: low- and medium-grade chimeras would indicate quality, recommend adding low/medium grade aneuploid/mosaic chimeras.

Fixed

(68) Line 725-729: it may be my mistake, but I think the results description is not found within the Results section, but only here in the legend? Please include this detail also in the Results section.

Fixed

(69) Line 729: which is AZ or Rev cells?

(70) References - Page number missing for some references; abbreviated version vs. non abbreviated version of journal titles used. Please be consistent/meet journal requirements.

Fixed

(71) Figures

Figure 1: [C] both AZ-NANOG and DMSO-SOX17 have mean/median(?) of 11 cells (described in results), yet in this figure (on the same axis) these groups are not level. Are the numbers correct? This is also the case for Rev-SOX17 which is described in the results as having 8 cells yet appears to be above the 8 mark in the graphs; AZ-CDX2, which has 64 cells yet appears to be below the 60 mark; AZ-total, which has 82 cells yet appears to be below the 80 mark. In [E] the label orientation, "ns" has both horizontal and vertical orientation. Please make appropriate changes throughout to reflect accuracy.

Figure 3: [C] As for Figure 1, DMSO-NANOG, which is described in results as having 14 cells, yet appears to be below the 13 mark in the graph; DMSO-SOX17, which has 6 cells yet appears to be above the 7 mark.

These is due to average

Figure 4: [D and E] random numerals appear in the bars on the graph. 9,10 and 7, 14? Are these sample size numbers? If they are, they should appear in all bars/groups or in the legend.

Yes, these are sample sizes

Figure 5: [D and G] same comment as for Fig 4 above, random numbers in the graph.

Yes, these are sample sizes

(72) Supplementary figures. Figure S2 [A] No quantification? This is important to add as representative images are only a 2D plane, which can be easily misinterpreted. [E] Should the y-axis label be written as "Number of cells normalised to DMSO group", or similar? Or is there a figure missing to depict the ratio of cells in each cell lineage normalised to the DMSO group, which is the description written in the legend? But I don't see a figure showing the ratio, just the absolute number of cells. Is this a missing figure or a mislabelled axis?

Quantification at the blastocyst stage is misleading due to high cellular heterogeneity.

Reviewer #3 (Recommendations for the authors):

(1) The statement in the abstract: "embryos with a low proportion of aneuploid cells have a similar likelihood of developing to term as fully euploid embryos" Line 48-50 Capalbo does not really answer as the biopsy may not be reflective of ICM.

This is a great point. Trophoctoderm biopsies may not reflect the real proportion of aneuploidy in the ICM. We emphasize this in discussion and Fig. S4.

(2) Line 69/70, at least 50% Singla et al/Bolton. It would be helpful to elaborate a bit more on this study. How can this be assessed when analysis results in destruction?

(3) Differences in the developmental potential of reversine versus AZ-treated embryos. It is not entirely clear why. The differences in non-dividing cells if any are small, and the - crest cells are rather minor also. Could these drugs have other effects that are not evaluated in the study?

Yes, specifically, reversine has been shown to have several off-targets effects. Including inducing apoptosis (Chen et al 2024).

(4) Lines 45-46 understanding of reduction of aneuploidy should mention/discuss the paper of attrition/selection, of the kind by the Brivanlou lab for instance, or others. As well as allocation to specific lineages, including the authors' work.

Dr. Brivanlou experiments in gastruloids do not represent the same developmental stage of pre-implantation embryos. Comparison between models is debatable.

(5) Line 53: human experiments are more limited due to access to samples. What does 'not allowed' mean? By who, where?

NIH does not allow to experiment with human embryos for ethical reasons.

(6) The figure callouts to S1A in lines 93,97. What is a non-dividing nucleus? For how long is it observed?

A non-dividing nucleus is an accumulation of DNA in a round form without define separation of the chromosomes and their specific kinetochores (CREST antibody). The presence of non-dividing nucleus during the 4 -to-8 cell stage can indicate activation of the spindle assembly checkpoint during prometaphase. Example of non-dividing nucleus can be observed in Fig S1.B.

(7) Line 108 A relatively minor effect on cell number and quality of blastocysts is observed. It is not surprising that thereafter, developmental potential is also high. At that stage, what are the individual cell karyotypes?

Due to technical limitations, we can't determine the specific karyotypes of these cells.

(8) Line 153. The p53 increase of 1.3 fold is not dramatic.

The levels of p53 at the morula stage is 7-fold differences. In contrast, at the blastocyst stage, a change in 1.3-fold is indeed less dramatic. This can be a result of the elimination of aneuploid cells or mechanism to counter the activation of the p53 pathway, like overexpression of the Hif1a pathway.

(9) Line 155. Is there a more direct way to test for p38 activation?

Natale et al 2004 (Dev Biol) and Sozen et al 2015 (Mech of Dev) described that inhibition of p38 deeply affect the development of pre-implantation embryos after the 8-cell stage. For this reason, comprehensible dissect the interaction between p53, HIF1A and p38 during aneuploid stress is challenging. We do not discard a double function of p38 during lineage specification and in response to DNA damage.

(10) Line 191/192 Low oxygen conditions, is this equal to hypoxia? What is the definition of hypoxia here? The next sentence says physiological. Is that the same or different?

Low oxygen can be defined as hypoxia. This varies from 2% to 6%. Our definition of hypoxia is 5% concentration of oxygen with 5% concentration of CO₂, taking into consideration the standard levels of oxygen in the IVF clinics. Physiological oxygen in mouse varies from ~1.5% to 8%.

(11) The question is whether there is something specific about HIF1 and aneuploidy, or whether another added stress would have similar effects on the competitiveness of treated cells.

That is a great follow up of our work.

(12) Line 300. Is p21 unregulated at the protein level or mRNA level? Please indicate.

mRNA level.

(13) *Figure 1D/E H2Ax intensity is cell cycle phase-dependent. It might be meaningful to count foci by the nucleus and show both ways of analysis.*

(14) *Check the spelling of phalloidin.*

Fixed in text and figures!

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